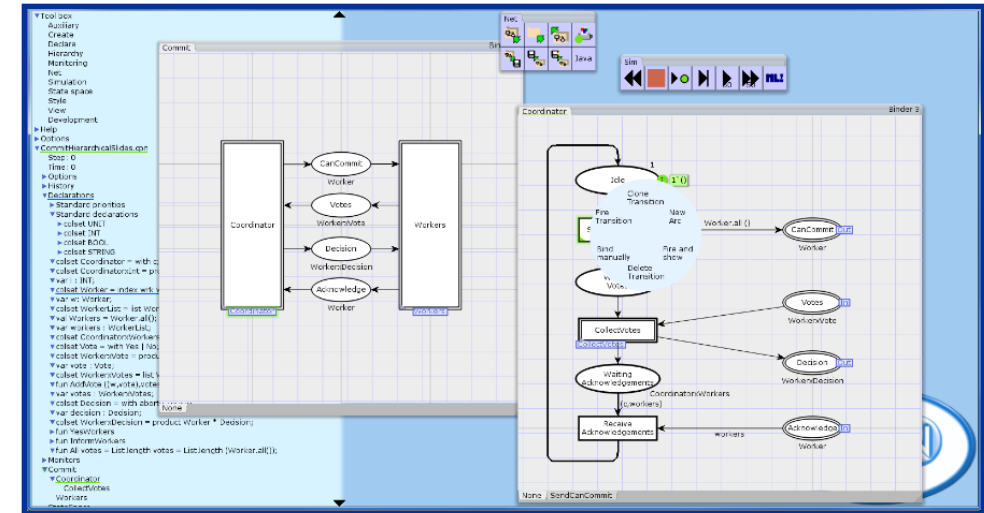


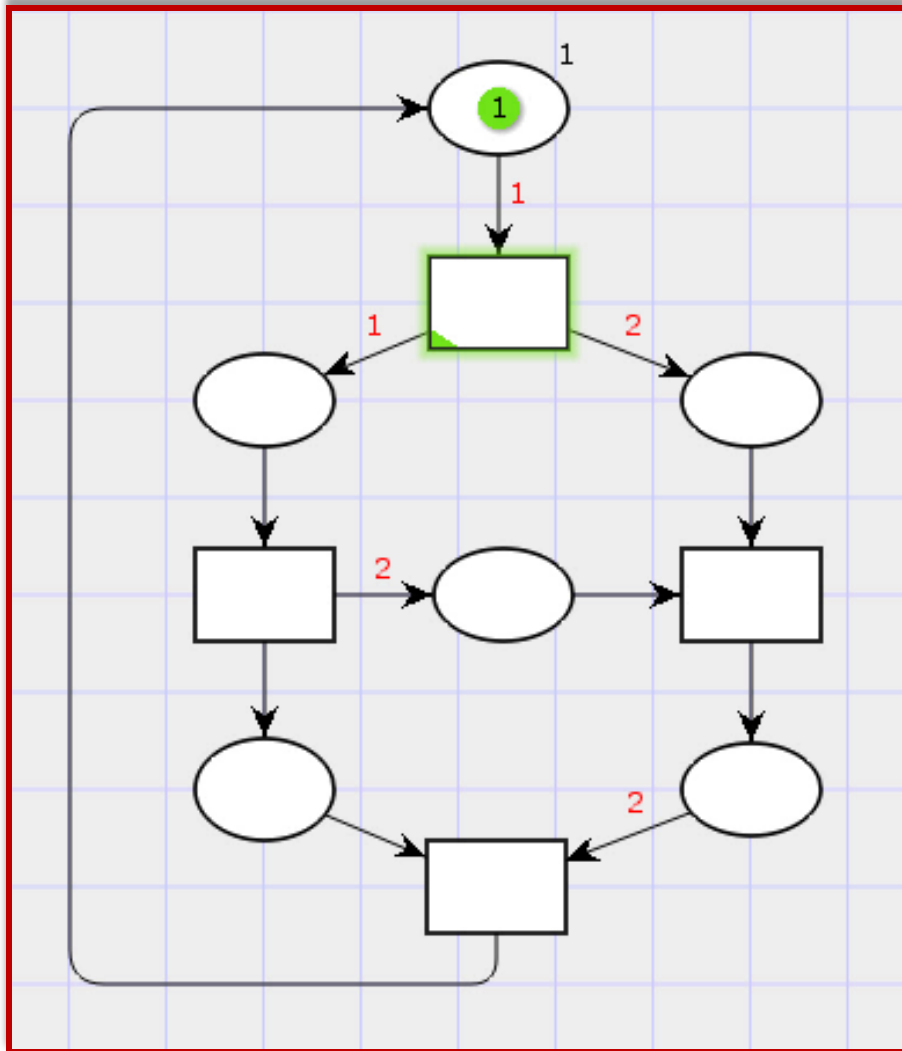
Background on Coloured Petri Nets

From Place/Transition Nets to Coloured Petri Nets



Lars Michael Kristensen
Department of Computer Science, Electrical Engineering, and Mathematical Sciences
Western Norway University of Applied Sciences

Quick Recap: Petri Net Concepts



State modelling

- **Places** (ellipses) that may hold **tokens**
- **Marking (state)**: distribution of **tokens** on the places
- **Initial marking**: initial state

Event (action) modelling

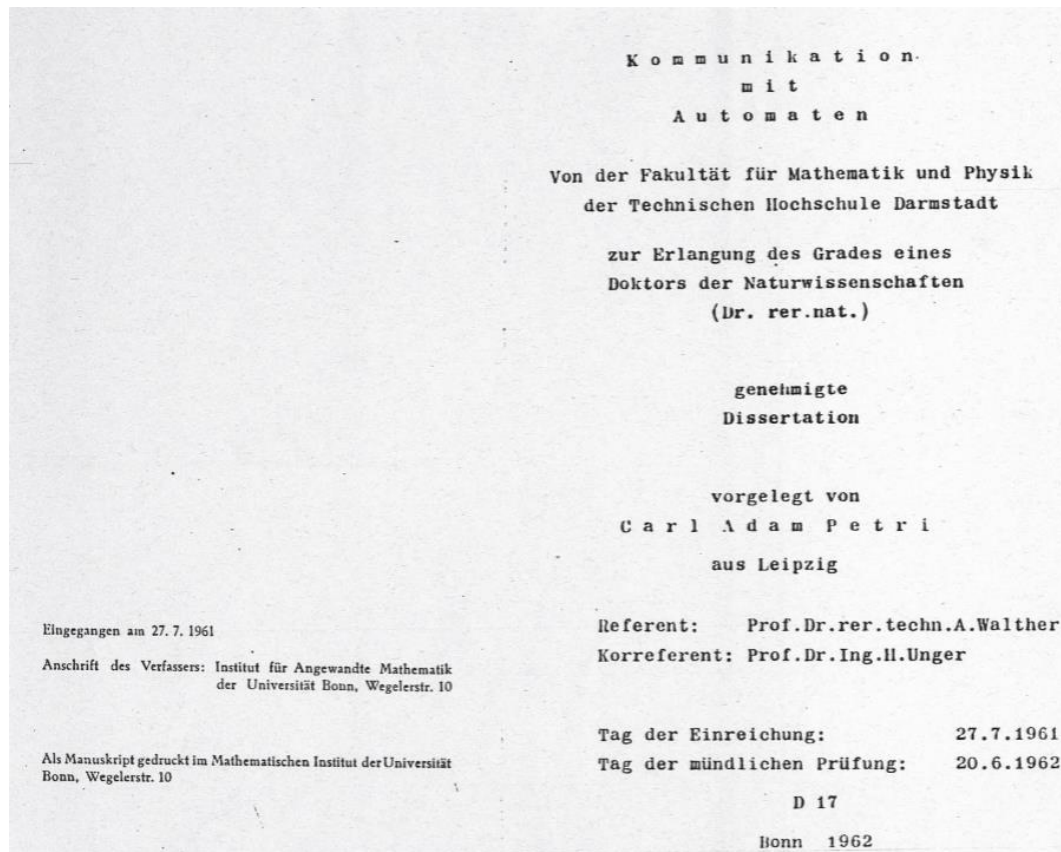
- **Transitions** (rectangles)
- **Directed arcs**: connecting places and transitions
- **Arc weights**: specifying tokens to be added/removed by transitions

Execution (token game)

- **Current marking**
- **Transition enabling**
- **Transition occurrence**

Petri nets

- Originates from the PhD dissertation of Carl Adam Petri (1926 – 2010)



High-level Petri Nets

- Petri Nets are divided into **low-level** and **high-level Petri Nets**
 - **Low-level Petri Nets** (such as Place/Transition Nets) are primarily suited as a **theoretical model** for concurrency - but are also applied for modelling and verification of **hardware systems**
 - **High-level Petri Nets** (such as CP-nets and Predicate/Transition Nets) are aimed at **practical use** and **software systems** - in particular because they allow for construction of compact and parameterised models
- High-level Petri Nets is an **ISO/IEC standard***
 - The CPN modelling language and the supporting CPN Tools conform to this standard

* <https://www.iso.org/standard/38225.html>

CPN models are formal

- The CPN modelling language has a **mathematical definition** of both its **syntax** and **semantics**
- The **formal representation is important**
 - Would have been impossible to develop a sound and powerful modelling language without it
 - Provides the foundation for the definition of the behavioural properties and for the formal analysis and verification methods

Definition 4.2. A non-hierarchical Coloured Petri Net is a nine-tuple $CPN = (P, T, A, \Sigma, V, C, G, E, I)$, where:

Definition 4.5. A step $Y \in BE_{MS}$ is enabled in a marking M if and only if the following two properties are satisfied:

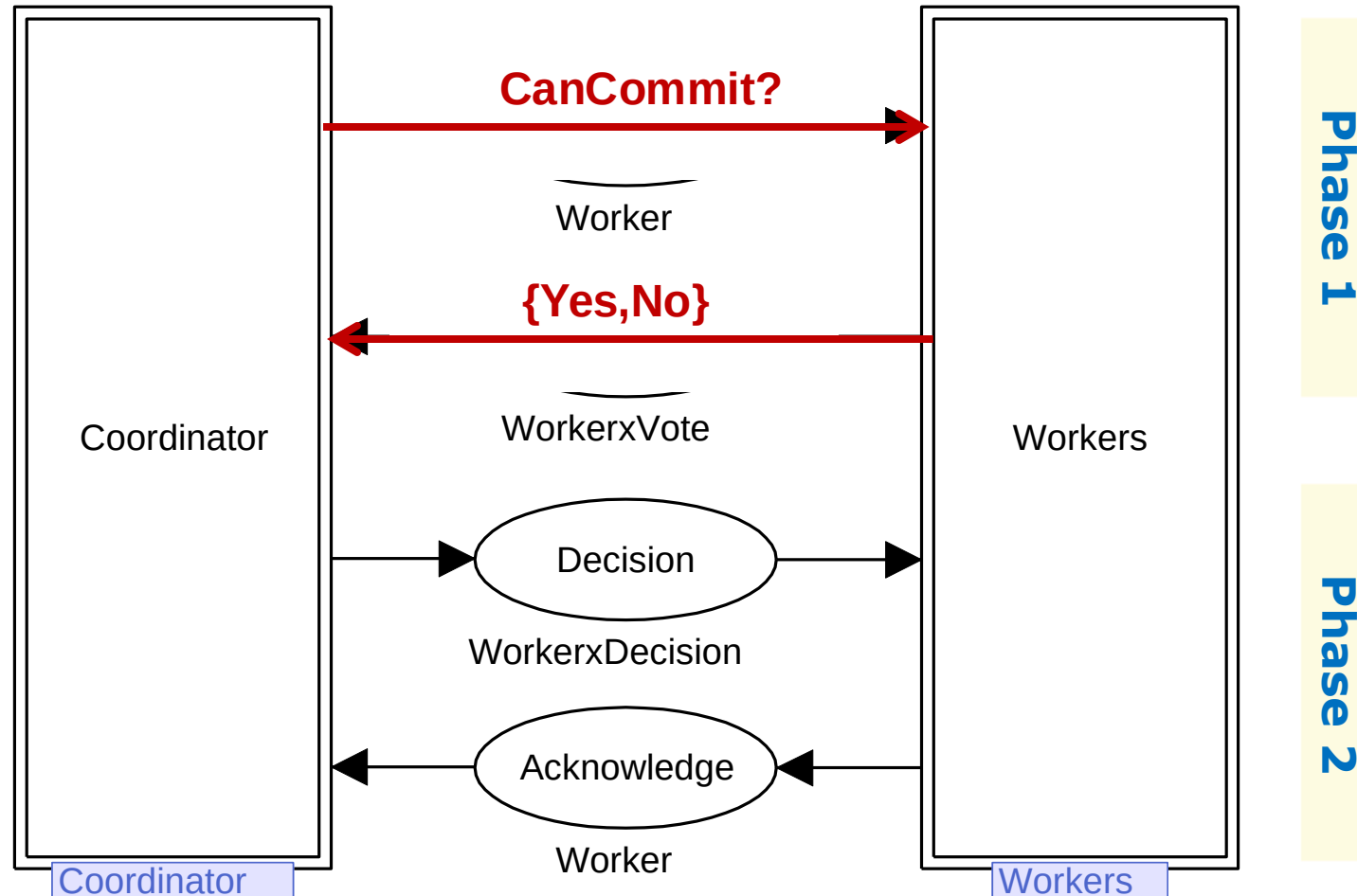
1. $\forall (t, b) \in Y : G(t) \langle b \rangle$.

Learning CPNs for practical use is similar to learning a programming language (no explicit mathematics :-)

8. $E : A \rightarrow EXP_{R_0}$ is an expression function that assigns an expression to each arc a such that $Type[E(a)] = C(p)_{MS}$, where p is the place connected to the arc a .
9. $I : P \rightarrow EXP_{R_0}$ is an initialisation function that assigns an initialisation expression to each place p such that $Type[I(p)] = C(p)_{MS}$.

Two-phase commit protocol

- How to model phase 1 with PT-nets?



CPN Tools – PT-net modelling

pt-tpc-empty.cpn | pt-tpc-complete.cpn

- **Construction, editing and simulation of Place/Transition Petri Net models in CPN Tools**
- **First part of the two-phase commit protocol using Place/Transition Nets**
 - How to model send and receive CanCommit with one worker?
 - How to model Yes/No votes?
 - How to model multiple workers?

